negative portion of the frequency spectrum is automatically included. The positive and negative frequencies are treated with an equal status; it requires one-half of each to form a complete waveform.

The Complex DFT

The forward complex DFT, written in polar form, is given by:

EQUATION 31-5
The forward complex DFT. Both the time domain, $x[n]$, and the frequency domain, $X[k]$, are arrays of complex numbers, with k and n running from 0 to N-1. This equation is in polar form, the most common for DSP.

$$X[k] = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}$$

Alternatively, Euler's relation can be used to rewrite the forward transform in rectangular form:

EQUATION 31-6
The forward complex DFT (rectangular form).

$$X[k] = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\left(\cos(2\pi kn/N) - j\sin(2\pi kn/N)\right)$$

To start, compare this equation of the *complex Fourier transform* with the equation of the *real Fourier transform*, Eq. 31-1. At first glance, they appear to be identical, with only small amount of algebra being required to turn Eq. 31-6 into Eq. 31-1. However, this is very misleading; the differences between these two equations are very subtle and easy to overlook, but tremendously important. Let's go through the differences in detail.

First, the real Fourier transform converts a real time domain signal, $x[n]$, into two real frequency domain signals, $ReX[k]$ & $ImX[k]$. By using complex substitution, the frequency domain can be *represented* by a single complex array, $X[k]$. In the complex Fourier transform, both $x[n]$ & $X[k]$ are arrays of complex numbers. A practical note: Even though the time domain is complex, there is nothing that *requires* us to use the imaginary part. Suppose we want to process a real signal, such as a series of voltage measurements taken over time. This group of data becomes the real part of the time domain signal, while the imaginary part is composed of zeros.

Second, the real Fourier transform only deals with *positive* frequencies. That is, the frequency domain index, k, only runs from 0 to $N/2$. In comparison, the complex Fourier transform includes both *positive* and *negative* frequencies. This means k runs from 0 to N-1. The frequencies between 0 and $N/2$ are positive, while the frequencies between $N/2$ and N-1 are negative. Remember, the frequency spectrum of a discrete signal is periodic, making the negative frequencies between $N/2$ and N-1 the same as